

Lab 3 report Jack Kane

To setup this lab I used MYSYS UCRT64 to download all the needed packages, (GCC, Cmake, Ninja and GDB) I then added MYSYS to my windows path

I then built my project by using the following commands:

```
cmake -S . -B build -G Ninja
```

```
PS C:\Users\gameuser\Desktop\GamesEng\GamesENGI\assignments\C00300324\Lab03> cd student_starter
PS C:\Users\gameuser\Desktop\GamesEng\GamesENGI\assignments\C00300324\Lab03\student_starter> cmake -S . -B build -G Ninja
```

```
cmake --build build --target lab03_tests
```

```
PS C:\Users\gameuser\Desktop\GamesEng\GamesENGI\assignments\C00300324\Lab03\student_starter> cmake --build build --target lab03_tests
```

I then ran the tests using the command:

```
.\build\lab03_tests.exe --gtest_color=yes
```

```
PS C:\Users\gameuser\Desktop\GamesEng\GamesENGI\assignments\C00300324\Lab03\student_starter> .\build\lab03_tests.exe --gtest_color=yes
Running main() from C:/Users/gameuser/Desktop/GamesEng/GamesENGI\assignments/C00300324/Lab03/student_starter/build/_deps/googletest-src/go
[=====] Running 5 tests from 1 test suite.
[-----] Global test environment set-up.
[-----] 5 tests from Calculator
[ RUN     ] Calculator.Add_ReturnsSumForPositive
[ OK      ] Calculator.Add_ReturnsSumForPositive (0 ms)
[ RUN     ] Calculator.Add_ReturnsSumForNegatives
C:/Users/gameuser/Desktop/GamesEng/GamesENGI\assignments/C00300324/Lab03/student_starter/test/lab03_tests.cpp:18: Failure
Failed
not implemented

[ FAILED  ] Calculator.Add_ReturnsSumForNegatives (2 ms)
[ RUN     ] Calculator.Subtract_ReturnsDifference
C:/Users/gameuser/Desktop/GamesEng/GamesENGI\assignments/C00300324/Lab03/student_starter/test/lab03_tests.cpp:23: Failure
Failed
not implemented

[ FAILED  ] Calculator.Subtract_ReturnsDifference (2 ms)
[ RUN     ] Calculator.Subtract_ReturnsZeroForEqualInputs
C:/Users/gameuser/Desktop/GamesEng/GamesENGI\assignments/C00300324/Lab03/student_starter/test/lab03_tests.cpp:28: Failure
Failed
not implemented

[ FAILED  ] Calculator.Subtract_ReturnsZeroForEqualInputs (2 ms)
[ RUN     ] Calculator.Add_HandlesZero
C:/Users/gameuser/Desktop/GamesEng/GamesENGI\assignments/C00300324/Lab03/student_starter/test/lab03_tests.cpp:33: Failure
Failed
not implemented
```

I then created the last 4 tests and rebuilt the project, after running it all 5 tests then succeeded

```

PS C:\Users\gameuser\Desktop\GamesEng\GamesENGIassignments\C00300324\Lab03\student_starter> .\build\lab03_tests.exe --gtest_color=yes
Running main() from C:/Users/gameuser/Desktop/GamesEng/GamesENGIassignments/C00300324/Lab03/student_starter/build/_deps/gtest-src/go
[=====] Running 5 tests from 1 test suite.
[-----] Global test environment set-up.
[-----] 5 tests from Calculator
[ RUN    ] Calculator.Add_ReturnsSumForPositive
[ OK     ] Calculator.Add_ReturnsSumForPositive (0 ms)
[ RUN    ] Calculator.Add_ReturnsSumForNegatives
[ OK     ] Calculator.Add_ReturnsSumForNegatives (0 ms)
[ RUN    ] Calculator.Subtract_ReturnsDifference
[ OK     ] Calculator.Subtract_ReturnsDifference (0 ms)
[ RUN    ] Calculator.Subtract_ReturnsZeroForEqualInputs
[ OK     ] Calculator.Subtract_ReturnsZeroForEqualInputs (0 ms)
[ RUN    ] Calculator.Add_HandlesZero
[ OK     ] Calculator.Add_HandlesZero (0 ms)
[-----] 5 tests from Calculator (7 ms total)

[-----] Global test environment tear-down
[=====] 5 tests from 1 test suite ran. (11 ms total)
[ PASSED ] 5 tests.
PS C:\Users\gameuser\Desktop\GamesEng\GamesENGIassignments\C00300324\Lab03\student_starter>

```

I did not run into any issues during this lab.

Using framework like this makes it easier to tell when anything goes wrong as the tests will detect it, this makes larger scale projects easier to manage as if anyone adds code that breaks as long as it has tests made for it the tests will start failing and it will be easy to see what went wrong allowing it to be corrected.